

Chen Tetro

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About Me

Computer Science Graduate with hands-on experience in full-stack software development , building web applications using React, Next.js, TypeScript, JavaScript, Node.js, and AI-powered architectures. Developed end-to-end projects including RAG systems, AI chat platforms, and scalable web applications. Strong foundation in software architecture, problem-solving, and modern development workflows.

Technical Skills

- **Languages:** C, C++, C#, Java, Python, JavaScript, TypeScript, Kotlin (Object-Oriented Programming) .
- **Frontend:** React, Next.js, HTML, CSS, Tailwind CSS.
- **Backend:** Node.js, Express.js, REST APIs.
- **Databases:** MongoDB, MySQL, Firebase, Supabase.
- **Tools:** Git, GitHub, Postman, CI/CD, GitHub Actions, Vitest, Cursor, Agile Workflows.

Projects:

Expense Management System

Tech: **React, Vite, Node.js, Express, MongoDB, Mongoose**

GitHub: <https://github.com/chentetro/expensemanagementsystem>

Demo: <https://expensemanagementsystem-frontend.onrender.com/login>

- Developed a full-stack expense tracking application to help users record spending and monitor financial patterns.
- Built REST API endpoints with Express and MongoDB for expense CRUD operations and client-server data sync.
- Created a modern frontend with React + Material UI implementing efficient State Management and Axios for a smooth user experience.
- Deployed the application and maintained source control/versioning with Git and GitHub.

Engineering Portfolio & AI Chat Platform

Tech: **Next.js, React, TypeScript, Tailwind, Vercel AI SDK, Vitest**

GitHub: <https://github.com/chentetro/newportfolio>

Demo: <https://newportfolio-sigma-seven.vercel.app/>

- AI-driven software development focused on advanced architecture, Software Engineering principles, and rigorous quality control.
- Strategic AI Planning: Constructed detailed Work Plans for every feature, utilizing Agile methodologies (segmented into predefined Phases and Steps) in collaboration with AI Agents to ensure full traceability.
- Rules & Commands Management: Implemented custom Cursor Rules (.mdc) and commands to enforce Folder-Role-Separation architecture, WCAG AA accessibility standards, and Clean Code principles.
- Agentic Code Review: Established a recursive feedback loop where AI serves as an active Code Reviewer, identifying logic, security, and syntax issues before component approval.
- Automation & Quality Assurance: Integrated Vitest to verify feature integrity (including the chatbot and UI components) in strict alignment with predefined work plans and architectural rules.

ActiveRAG-HIT (Under development)

Tech: **Next.js, TypeScript, Supabase, Google AI studio, Vercel AI SDK, LangChain,**

Tailwind CSS, shadcn/Base UI

GitHub: <https://github.com/chentetro/rag-hit>

Demo: [HIT Assistant](#) | [RAG Chatbot](#)

- Built a production-ready Hebrew-language AI academic assistant prototype designed to support HIT students using Next.js and TypeScript.
- Engineered a complete RAG pipeline by crawling, chunking, and embedding institutional website content into a Supabase vector database for highly accurate search.
- Designed a responsive frontend architecture featuring a real-time streaming chat UI, automated source citations, and robust mock environments to simulate user-scoped tools and bypass organizational access restrictions.
- Implemented CI/CD pipelines and automated backend workflows by configuring GitHub Actions for scheduled recrawling and continuous integration.

Education

B.Sc. Computer Science, HIT

GPA: 90 Data Structures & Algorithms | Object-Oriented Programming | Operating Systems | Database Systems | Software Engineering | Computer Networks

Full High School Diploma.

Experience

2024-2025:

Administrative Assistant – Head of Quality Control Division, Elta

- Managed calendar, scheduling, and administrative tasks .
- Coordinated with HR and organized team-building activities .
- Supported project workflows and documentation .

2022-2023:

Military Service, Air Force Headquarters – Career Service

- Handled logistical requests and procurement with external vendors .
- Drafted official documents and supported cross-team projects .
- Gained experience in task prioritization and decision-making under pressure .

2020-2022:

Military Service, Air Force – Regular Service

Diagnosed and resolved mission-critical avionics failures under strict operational constraints, requiring analytical troubleshooting, precision, and rapid decision-making.

Languages

Hebrew: Native

English: Fluent